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# From Eastern Philosophy to Craft and Innovative Education: A Study on Practical Implementation

**Yun-Chi Lee [1,2,*] and Tii-Jyh Tsay [2]**

1 Doctoral Program in Design, College of Design, National Taipei University of Technology, No. 1, Sec. 3, Zhongxiao E. Rd., Da'an Dist., Taipei City 106344, Taiwan

2 National Taiwan Craft Research and Development Institute, No. 573, Zhongzheng Rd., Caotun Township, Nantou County 542020, Taiwan; tii@ntcri.gov.tw

* Correspondence: yunchilee0604@gmail.com; Tel.: +886-933-561-741

**Abstract:** This study explores the application of Eastern philosophy in craft innovation education, identifying opportunities for interdisciplinary learning. Drawing on the *I Ching* and Laozi's thought, it examines human needs in craft across three dimensions: Qi-form (material), Xin-form (psychological), and Dao-form (philosophical). Taiji theory's Yin–Yang balance highlights the importance of interdisciplinary thinking in craft innovation. This study introduces the "Spiral Innovation Theory" as a framework for craft education, implemented in the 2024 Taiwan Craft Academy Summer Program with 43 participants. The curriculum covered lacquer, wood, metal, and ceramics, employing a multi-mentor system. Using the Learning Motivation Strategies Scale, Imaginative Thinking Scale, and interviews, the findings reveal that different crafts foster distinct creative abilities. The ANOVA results show woodworking enhances ideation, metalwork and ceramics improve fluency, ceramics and woodworking strengthen flexibility, while woodworking and lacquer work boost creativity. A significant correlation between learning motivation and imagination was found. These findings offer insights into future craft education, advocating the dual mentorship model as a strategy for interdisciplinary innovation.

**Keywords:** The Book of Change; crafts; interdisciplinary education; imaginative thinking

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## 1. Introduction

Eastern philosophical thought emphasizes the values of harmony between humans and nature and adherence to the natural order. These concepts have long influenced Eastern lifestyles, art, crafts, and even the overall social structure. For instance, Laozi places particular importance on the idea of "wu wei" (non-action), which stresses that humans should respect the laws of nature and avoid excessive intervention, thereby achieving harmony and balance [1,2]. The *I Ching* (*Book of Changes*) meticulously discusses the transformation and change of things, exploring the laws of the universe through the observation and understanding of Yin and Yang and the five elements (metal, wood, water, fire, earth), inspiring people to seek wisdom in adapting to the environment and embracing change in both life and work. These understandings of change and adaptation resonate closely with the core principles of contemporary design thinking, especially when facing complex challenges such as globalization and environmental changes. Eastern philosophy thus provides a crucial perspective for rethinking design and craft innovation.

Amidst the rapid mass production and fast fashion consumption driven by ecological transformations, traditional craft techniques are increasingly being replaced by mass-produced, standardized technologies in product development. In response to market preferences, some designers opt for non-natural craft techniques, inadvertently diminishing the inherent advantages of designs rooted in Eastern philosophical traditions. Craft, as an important cultural expression and practical skill in human life, is not only the transmission of techniques but also the continuation of culture and philosophy, which is conveyed through tangible creative works that express deeper meanings [3]. In contemporary design education, the importance of craft innovation education should be further emphasized. This study seeks to explore the possibilities of craft innovation education through Eastern philosophical thought, aiming to integrate traditional philosophy with modern design thinking. By merging craft techniques with sustainability principles, craft innovation can become a key approach to addressing contemporary design and environmental challenges.

After extracting the theory of craft innovation education from traditional Eastern philosophical thought, this study will collaborate with the National Taiwan Craft Research and Development Institute (hereinafter referred to as the Craft Center), located in central Taiwan, to develop a craft innovation education camp. The Craft Center, affiliated with the Ministry of Culture, is the highest authority for promoting Taiwan's craft industry and education. It was established in 1954, during the post-World War II era, when handcraft products were being exported internationally. After 2000, Taiwan's craft development entered the era of knowledge and digital economies, enhancing the value of crafts through technology-assisted design innovation and digital applications. In 2002, the government made the "Cultural and Creative Industries" a key policy focus, with the craft industry becoming an important role in the new cultural economy. In 2010, the Taiwan Craft Research Institute was upgraded to the Craft Center, promoting craft revival policies and establishing several branches in Taiwan to innovate, research, apply, and promote traditional crafts while cultivating a new generation of craft design talents.

This study collaborates with the Craft Center to organize the 2024 Taiwan Craft Academy summer talent training camp and initiates a planning strategy coordinated by the leaders of various craft laboratories at the Craft Center, inviting university students from Taiwan to participate. We aim to find balanced opportunities for contemporary education from traditional Eastern philosophical thought and place particular emphasis on the practical action of craft innovation education. The objectives of this study are summarized as follows:

1. By reviewing Eastern philosophical thought, propose a theory for craft innovation education.
2. Implement the "Craft Innovation Education Theory" into the project strategy and curriculum design planning for the 2024 Taiwan Craft Academy summer talent training camp while focusing on students' imaginative thinking performance during their craft learning process.
3. Reflect on traditional Eastern philosophical thought as a potential for contemporary craft innovation education.

## 2. Literature Review

### *2.1. Eastern Philosophical Thought*

Classics of traditional Chinese philosophy include the *I Ching*, *Book of Documents*, *Book of Odes*, *Book of Rites*, and *Spring and Autumn Annals*, among which the *I Ching* is particularly famous for its natural and humanistic philosophical ideas. As a core text of Chinese culture, the *I Ching's* concept of "shengsheng zhi wei yi" (continuous change is the essence of the *I Ching*) emphasizes the natural law of constant change between heaven and earth. This change is vibrant and serves as the source of growth and innovation for all things [4]. The aesthetic philosophy of the *I Ching* values nature and law, with its key points including "the Way of Heaven is Yin and Yang; the Way of Earth is flexibility and rigidity; the Way of Humanity is benevolence and righteousness". These viewpoints stem from observing the harmony of Yin and Yang and the seasonal changes in nature, revealing the coordination and interaction between movement and stillness, expansion and contraction, large and small, and hardness and softness. It also presents principles for human conduct, such as benevolence, justice, propriety, and morality, forming an aesthetic philosophy of "respect for life, and harmonious regulation" [5]. Laozi's Daoist thought also emphasizes "Dao follows nature" and advocates that humans should align with the natural order and govern through non-action. This idea is similarly significant for craft design education [6]. In modern educational practice, this perspective on nature requires educators to guide students in understanding the beauty of nature, drawing inspiration from it, and creating craft works that adhere to natural laws. This is not merely the development of technical skills but is a process of discovering and creating, through nature, a profound experience and practice of the virtues of heaven and earth.

The *I Ching's* Xi Ci II states: "There is Taiji in the I Ching, which produces the two opposites. The two opposites produce the four symbols, the four symbols produce the eight trigrams, and the eight trigrams determine good and bad fortune, which leads to great achievement" [7]. This passage explains the origin of all things, the changing of the seasons, and natural phenomena, emphasizing the need to follow natural laws, which is "the Dao", also referred to as "Taiji". This viewpoint echoes Laozi's description in Chapter 42 of the *Dao De Jing*: "The Dao produces One, One produces Two, Two produces Three, and Three produces all things" [8]. In the "Three Levels of Needs Theory" Chen, Tian-Li & Zhuang, Yu [1], Chinese philosophy is presented as a form of life science, advocating the understanding of life through experiential living and returning to nature. In human interactions, one should follow natural principles to achieve "harmony between Heaven and Humanity". Such Eastern thought reminds designers and craftsmen that they should learn from the beauty of Yin and Yang, the beauty of flexibility and rigidity, to cultivate their own benevolence and righteousness, the so-called Three Ways. This philosophy not only explores the daily life of each individual from a micro perspective, ranging from basic survival needs to living needs to higher-level life needs, but also relates to the realization of "the Way of Craft", corresponding to form, spirit, and ultimately the shape of the Dao. This process describes the journey of creators and craftsmen, from technical execution by hand to self-understanding and eventually to the pursuit of the Dao. The inner world manifests in the external world and social norms, constituting the operation of social order, which is referred to as the "great achievement" or, in Buddhist terms, the "Great Thousand World" (Figure 1). As the *I Ching's* Xi Ci II states: "Great achievement is wealth; flourishing virtue is renewed daily; continuous change is the I Ching; the formation of symbols is Qian (the creative); the imitation of the Dao is Kun (the receptive)". Thus, humans and all things will live harmoniously, achieving sustainable development and reaching the realm of "good" and "eternity". This Eastern philosophical thought has, in fact, anticipated what is now emerging as the contemporary concept of sustainability!

This study aims to explore the practical possibilities of applying Eastern philosophical thought through action research. It is hypothesized that an education philosophy based on Eastern philosophy can help cultivate students' comprehensive qualities, inspire confidence in traditional culture, and enhance their innovative capabilities, thereby achieving a deep integration and development of education and culture [9,10].

### *2.2. Innovation from a Cross-Disciplinary Perspective*

In *I Ching's* Hexagram Feng, it states, "When the sun reaches its zenith, it begins to decline; when the moon is full, it starts to wane". This describes the changes and constants in the natural world, as well as the rhythm and movement of all things. From an aesthetic perspective, as previously mentioned, the *I Ching* emphasizes the importance of the timely coordination of Yin and Yang, which naturally forms a harmonious aesthetic philosophy of the unity of opposites. Craft items, as functional objects, not only seek beauty in their design but also aim to convey the creator's thoughts and spirit. This corresponds to the contemporary design field's discussions on collaborative design. Collaboration is commonly found in various design disciplines, and scholars such as Kwan and Ofori [11] and Butterfoss [12] argue that a collaborative design must be based on the individual collaborators' interests. This allows each collaborator to share their skills, power, and resources while respecting the values of others, thereby contributing their professional abilities towards a common goal [13–15].

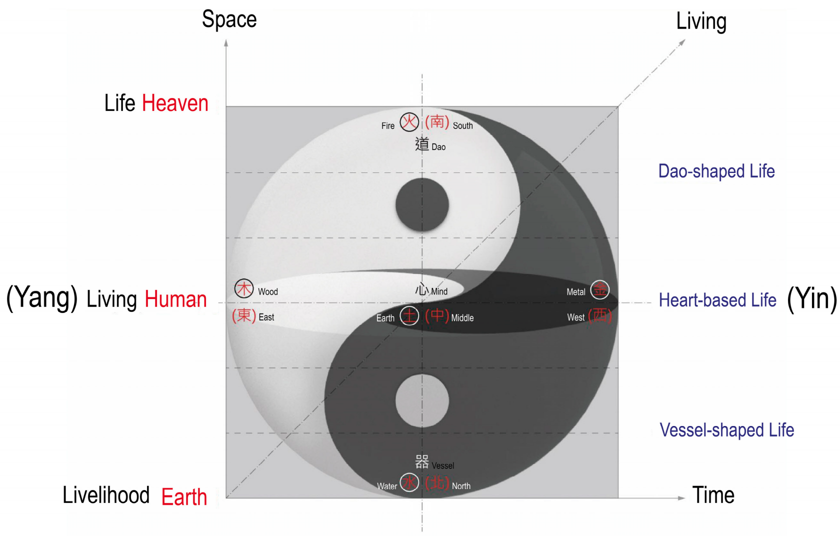


**Figure 1.** The Three-L Demands Theory [1].

Modern design perspectives are aligned with philosophical ideas. In the *I Ching*, the sixty-four hexagrams represent different combinations of hexagrams, illustrating the relationship between "time and place". A suitable combination of time and place can lead to favorable outcomes, while an incorrect combination results in unfavorable ones. From a human perspective, this is similar to the idea of "placing the right person in the right position" in collaborative management, which ensures that each participant's strengths are maximized. At a higher level, collaboration may not always involve "individuals" as units; if various individuals can achieve mutual respect, appropriate contributions, and compromise, it corresponds to the *I Ching* (Xici Xia) saying, "Fortune and misfortune are determined by steadfastness. The Way of Heaven is steadfast observation. The Way of the Sun and Moon is steadfast brightness. The movement of the world is steadfast unity". This explains the ebb and flow of Yin and Yang, the life and death cycles, and the natural changes in the universe. The concept of Taiji (the Great Ultimate) in both philosophy and its symbolic representation in crafts inherently involves the notion of "cross-disciplinary" and "harmony".

From a craft perspective, cross-disciplinarity could refer to the use of color, material, tools, or, more likely, diverse elements of thought and concepts. The timely adjustment and respect for changes between elements can explain the harmonious collaboration of cross-disciplinary efforts. A more modest and respectful attitude allows the elements to cooperate, symbolizing Taiji. The harmony of all things does not result in stillness but rather symbolizes the vibrant and sustainable potential of life. Hooimeijer et al. [16], in their study on interdisciplinary design, similarly argue that the process of interdisciplinary collaboration integrates previously independent knowledge domains into an innovative system of shared value. This suggests that interdisciplinary harmonization does not lead to stagnation; rather, it symbolizes vitality and growth, becoming a continuous and sustainable source of innovation. As the *I Ching* (Xici Xia) says, "The interaction of hardness and softness brings change within". It shows that the growth process of all things involves the fusion of Yin and Yang, forming Qi in the process, which represents the origin of life. The relationship between "change" and "constancy" is referred to as the "Dao" in Laozi's philosophy. From the concept of Taiji, the inevitable relationship between cross-disciplinary fields can be explained (Figure 2). In creative processes, the contemplation of the qualities of change and constancy, along with creation that harmonizes with "Qi, time, and change", results in a fusion of the present and the past, driving innovation toward the future. This is a critical perspective on craft aesthetics, as reflected in the *I Ching* (Bi) philosophy: "Observe the patterns of the heavens to perceive the changes of time; observe human patterns to transform the world". Exploring ancient Eastern philosophical thought aligns with the trend of contemporary interdisciplinary education, as such interdisciplinary competence is now regarded as a core ability for solving real-world problems [17]. Furthermore, scholar Ren [18] emphasizes that exploring traditional core cultural ideas will help people face and resolve various contemporary issues, including the balance of current problems in craft and applied arts education.

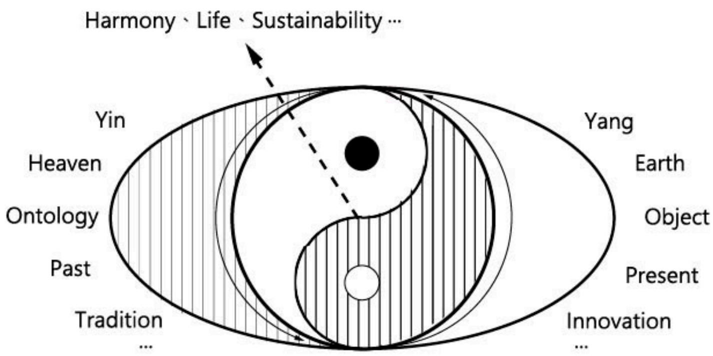


**Figure 2.** Cross-domain and harmonization relationship diagram (Data source: Created by NTCRI).

### 2.3. Apprenticeship and Craft Education

The scholar Pestalozzi (1746–1827) was the first to incorporate practical training into general education and advocated for the principles of manual education. He believed that manual labor is not only a physical activity but can also promote intellectual development. Through practical training, students can develop thinking abilities and apply these skills to their senses and muscles, ultimately mastering techniques and craftsmanship [19]. In early Taiwan, the traditional apprenticeship system was widely used as the primary mode of skill transmission in fields such as food, clothing, and architecture [20]. However, with the process of modern industrialization, mechanized technology gradually replaced many traditional handcrafted crafts, leading to the division of labor into skilled, semi-skilled, and unskilled work, which weakened the traditional craft hierarchy [21]. The shift in product manufacturing processes brought about by rapid industrialization turned the traditional 5–7-year apprenticeship into a learning constraint. In its place, the specialization of skills and the university education model gradually created a distinction between cognitive-based and hands-on work education institutions [22,23]. For example, in Taiwan, there is a division between senior high schools, vocational schools, universities, and technical universities, each corresponding to different levels of experience required for specific jobs, reflecting the changes and challenges in craft learning methods.

Nevertheless, in certain craft and labor-intensive industries in Taiwan, the one-on-one apprenticeship method remains, aiming to transmit craft expertise and bodily memory through the guidance of a master [24]. Craftspeople such as potters, blacksmiths, and carpenters are not only skilled technicians but also combine theory and practice. Through their daily work, they pass on technical knowledge and develop a deep technical foundation [24]. In the apprenticeship relationship, the master demonstrates professional attitudes, values, and skills in the workplace, while the apprentice gradually develops respect for and responsibility toward the work through this practice [25,26]. An apprenticeship not only helps improve the apprentice's technical skills but also has a positive impact on their job satisfaction and organizational commitment. A study on apprenticeships by Liu et al. [27] found that the apprenticeship system effectively reduces work stress and enhances apprentices' professional pride and job involvement. Noe [28] further pointed out that the quality of the apprenticeship program is positively correlated with apprentices' job satisfaction, suggesting that modern craft education should seek a balance between tradition and innovation. This line of thought may be illustrated in Figure 2, where this study attempts to interpret the philosophy of Yin–Yang and Taiji as a specific practice for cross-disciplinary collaboration and harmony in craft education. Such collaboration could involve cross-disciplinary techniques, categories, traditional and innovative integration, and conceptual exchanges. These could serve as opportunities for designing strategies in craft innovation education. Traditional craft education emphasizes the transmission of the "tacit knowledge" inherent in craft through hands-on experience and participation in the apprenticeship system [29,30]. Such a learning process will encourage learners to develop greater insight, imagination, and flexibility to tackle various challenges [31]. Of course, the advantages of the apprenticeship model are not only reflected in the improvement of skills and knowledge. It has been proven that the supportive behaviors, inspiration, appreciation, encouragement, and care provided by mentors during the educational process can enhance learners' motivation, imagination, and self-efficacy [32].

### *2.4. Learning Motivation*

It is unquestionable that a student's learning motivation will directly affect their learning outcomes. Numerous studies on students' short-term experiential learning have found that when students enter an activity with a strong desire to enrich their abilities, foster creative thinking, learn new skills, enhance self-efficacy, and believe that these learnings are related to their long-term future development, it maximizes the learning benefits [33–35]. Moghaddam [36] found, in a study on students' personality traits, that students with higher internal control, who enjoy learning new skills and face challenges, significantly improve their learning outcomes through self-managed learning decision-making behaviors [37]. Schunk [38] also supported this in his research on learning motivation, self-regulation, and the use of learning strategies. He argued that learning motivation is positively correlated with a student's self-discipline, mastery of learning goals, self-motivation, and initiative.

Regarding students' imagination and creativity in their creations, learning motivation also plays a crucial role. Amabile conducted a series of studies from 1979 to 1988 on motivation orientation and innovation indicators, proving that there is a strong causal relationship between a student's learning motivation and the imagination they can exhibit. Amabile further noted that students' learning motivation could be divided into intrinsic and extrinsic motivation. Although external pressures may enhance motivation intensity, they may harm students' imagination [39,40]. For measuring learning motivation, the Motivated Strategies for Learning Questionnaire (MSLQ), developed by Pintrich in 1986, provides an excellent tool for assessing students' motivation orientations and learning strategies. To explore the impact of learning motivation on students' effectiveness after participating in craft innovation education, this study will use this tool.

### 2.5. Craft as Imaginative Education

The most distinctive feature of craftsmanship lies in the hands-on bodily experience that brings advantages to the learning process, particularly supported by Dewey's theory of experiential learning. Dewey believed that learning arises from practice, and learners' experiences in real-life situations form the foundation for knowledge construction [41]. In craft education, students engage in hands-on activities, not only learning technical skills but also reflecting during the process, gradually deepening their understanding of craft culture and making craft skills, knowledge, and emotions part of their bodily memory [24]. Scholar Kolb's [42] experiential learning model further emphasizes that learning is a cyclical process, including concrete experience, reflection, abstract conceptualization, and active experimentation. Kolb [42] emphasized the interaction between learning and the environment, where learners are not passive recipients of information but construct knowledge through interaction with the environment. Vygotsky [43] argued that learning is a product of the socio-cultural context, and knowledge construction relies on social interaction and collaboration. During the practice, students, through interaction with teachers and classmates, share experiences, forming a social learning paradigm, which in turn promotes the socialization of knowledge and the transmission of culture. Therefore, we could even argue that during the teaching process, a form of cross-domain collaboration and harmonization emerges between the master and the apprentice, and individuals also engage in a close dialogue with the larger society. Ultimately, the technical, conceptual, attitudinal, and intellectual elements fuse into creative works, which are presented to the world, forming an innovative dialogue and content. It is understandable that this dialogic process is inherently bidirectional, allowing both masters and apprentices to achieve self-improvement through interaction and dialogue [44]. Ultimately, their techniques, concepts, attitudes, and understanding are integrated into creative works that are presented to the world, forming an innovative bridge of dialogue and meaning.

Walker and Unterhalter [45] pointed out the importance of reflective learning in enhancing individual imagination. Craft learning and creation often require the tempering of time, a process in which learners reflect on their creations to improve both their skills and creative thinking. Vygotsky [43] emphasized that creative activity arises from individual needs and desires through the reconstruction of past experiences to create something new. Repeated reflection allows creators to develop a deeper understanding of past knowledge and experiences, breaking away from the superficial learning model often found in traditional education [46]. Imagination in craft education is based on this principle, where students apply the techniques they have learned to new creations through experimentation and exploration, developing unique styles and creativity in the process. This creative activity is referred to as imagination in psychology [43]. A creator's reflection is inspired through observations and interactions with their environment and others. Deaux et al. [47] further suggested that observation, imitation, and interaction with others are ways in which learners reflect and construct their learning paradigms.

Chen, T.L., Hong, Y.H., and Lee, Y.C. [48], based on Eastern philosophical ideas, proposed a Spiral Innovation Theory to explain the overall innovation in craft. This study further integrates and adjusts the Taiji concept from the interdisciplinary and harmonization relationship diagram in Figure 2. It suggests that the Yin–Yang Taiji philosophy, which holds that all things are generated from the interplay of Yin and Yang, can serve as a crucial foundation for the Spiral Innovation Theory. This concept, like the nourishing nutrients for the growth of a tree that supports craft, can be seen as an invisible yet present implicit world within the human heart. This inner world resides in the creator, learner, educator, and others and also contributes to the formation of a collective social consciousness. The learner's moral enhancement of the inner world then follows the upward spiraling process from the "form of the object" and "form of the heart" to the "form of the way" (Figure 1). The spiral represents the experiences of trial, learning, and collision, gradually gaining clarity of the Dao (the Way) from the chaotic inner world and expressing this moral understanding through craft objects. As described by Chen, T.L., Hong, Y.H., Lee, Y.C. [48], through the nourishment of the soil, the "trunk" of the craft forms, and the "branches and leaves" of artistic and design expressions grow. Ultimately, from chaos, the logical system of the Dao emerges, and every craft tree develops its unique style in art and design expression, which is the manifestation of imagination. In other words, imagination is not merely an intellectual expression; more importantly, it is the creator's realization of the Dao, the crystallization of central ideas and imagination [49–51]. When all crafts come together, from point to line and from line to plane, they move toward the depth of knowledge, thus forming the visible, external world (Figure 3).

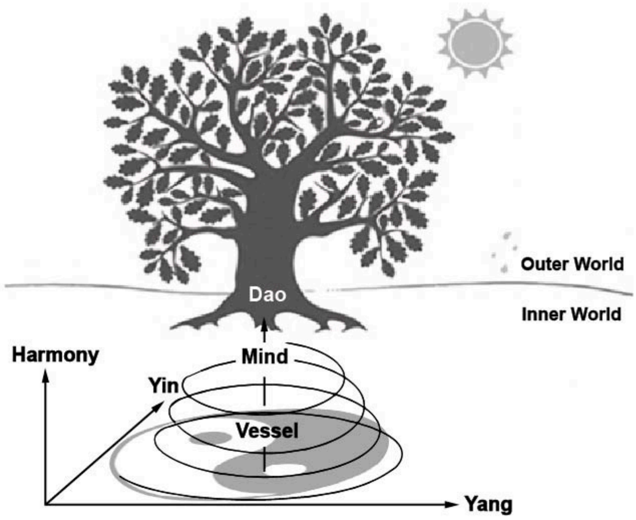


**Figure 3.** Spiral elevation of creativity (Data source: Modified from [48]).

## 3. Research Methods and Practical Strategies

### *3.1. Implementation Plan*

This study adopts an action research methodology as the strategy, based on the "Plan-Action-Observation-Reflection-Modification" framework to design the teaching content and detailed plan for the "2024 Taiwan Craft Academy summer talent training camp". The "2024 Taiwan Craft Academy summer talent training camp" is coordinated by the heads of various craft laboratories at the Craft Center, and the planning is based on the cross-disciplinary, collaborative, and harmonizing thinking principles mentioned earlier. The curriculum design includes cross-domain skills, collaborative teaching and creation, and integrative thematic design, with a particular emphasis on the integration of digital technology and its practical applications. The aim is to create a collision and harmonization process with traditional craft techniques, stimulating creativity and enhancing learning outcomes. The "2024 Taiwan Craft Academy summer talent training camp" has completed the recruitment of four types of laboratories—lacquer, wood, metal, and ceramics—where the experts include university faculty, senior artisans, product designers, and technology specialists. After the course content is planned, external experts are invited to review the curriculum to enhance teaching effectiveness. The course content is outlined in Table 1, which includes not only professional technical training but also aesthetic and artistic general education, specialized creative courses, and inter-laboratory exchanges. The course implementation begins with an introduction to Eastern philosophical theories, which are then applied to craft innovation learning. In the dual mentorship teaching environment, the instructors serve not only as transmitters of knowledge and skills but also as facilitators who guide students through reflective learning interactions. A reciprocal feedback approach is employed to enhance course management. At the end of the program, the participants must present their learning experiences and creative concepts at a public exhibition, ensuring a structured and rigorous learning process. In terms of research methodology, a mixed-methods approach is adopted to overcome the limitations of single-method studies. Qualitative data collection is particularly valuable in interpreting the quantitative findings. For the quantitative component, the Motivated Strategies for Learning Questionnaire (MSLQ) and the Imaginative Thinking Scale are used for data collection. The qualitative component consists of in-depth interviews, which explore how participants integrate Eastern philosophical thought into their creative processes, apply interdisciplinary thinking, and engage in hands-on craft innovation. The overall course implementation and research process are structured as illustrated in the following framework (Figure 4):

**Table 1.** "2024 Taiwan Craft Academy summer talent training camp" curriculum content for the four laboratories.

| Group | Course Title | Learning Objectives | Course Plan and Instructors | Hours |
|---|---|---|---|---|
| Lacquer Craft: | Lacquer Art Wearables 1: 3D Jewelry Creation Workshop | 1. Traditional lacquer art combined with contemporary media craft creation through 3D printing and traditional rare lacquer techniques in multi-material creative workshops<br>2. Broaden students' creative concept construction by utilizing the fusion of craft techniques in lacquer art and the inheritance of traditional lacquer art. | Experts and Instructors: Mr. Zhou and Mr. Wang Lacquer Craft Laboratory Supervisor: Ms. Huang | 108 h |
| Woodcraft | Woodcraft Combined with Digital Manufacturing Technology Application Training Course | 1. Forming and crafting furniture using computer software and digital manufacturing (CNC).<br>2. Learning the theory and practical operation of woodworking skills. | Experts and Instructors: Ms. Deng, Mr. Huang, and Mr. Wang Woodcraft Laboratory Supervisor: Mr. Jiang | 108 h |
| Metal Crafts | Between: A Dialogue Between Tin and Vase | 1. Exploring the processes of tin craft techniques and the relationship between tin and floral vessels.<br>2. Cultivating professional talent in contemporary craft design and aesthetics.<br>3. Establishing basic knowledge of tin art and concepts of floral vessel design. | Experts and Instructors: Ms. Xu, Mr. Lu, Ms. Wei, and Mr. Chen Head of the Metalworking Laboratory: Mr. Li | 108 h |
| Ceramic Arts | Composite Ceramic Sound and 3D Printing Design Course | 1. Exploring the processes of composite media craft techniques and their relationship with sound systems.<br>2. Cultivating professional talent in contemporary craft design and aesthetics.<br>3. Establishing basic knowledge in ceramic and wood craft. | Experts and Instructors: Mr. Zhang, Mr. Xu, Mr. Chen, and Mr. Liang Head of the Ceramic Arts Laboratory: Mr. Li | 108 h |

Source: 2024 Taiwan Craft Academy summer talent training camp recruitment brochure and official website.

### 3.2. Participants

The "2024 Taiwan Craft Academy summer talent training camp" primarily recruited students from universities and colleges across Taiwan. The Craft Center facilitated the promotion through official documents to relevant departments in these institutions. After screening for eligibility, a total of 43 participants successfully completed the training, consisting of 12 males and 31 females. The participants came from 26 universities and colleges across Taiwan, representing 31 different academic departments, including 11 from craft-related departments and 32 from cultural and creative design-related departments.

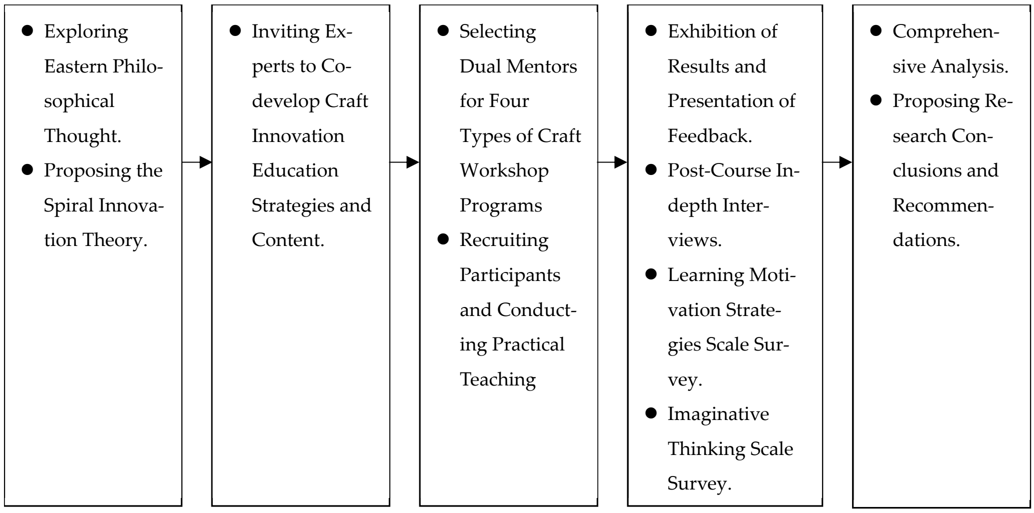


**Figure 4.** Explore Eastern philosophical thought.

### *3.3. Records and Analysis Tools*

#### 3.3.1. Learning Motivation Strategy

In research related to the student's choice to participate in off-campus internship activities, learning motivation is considered a key factor that influences students' learning outcomes.

For the exploration of learning motivation strategies, this study selects the Motivated Strategies for Learning Questionnaire (MSLQ), developed by Pintrich [52], which is primarily used to measure students' motivation orientation and learning strategies. In the 1993 version of the MSLQ, proposed by Pintrich, Smith, Garcia, and McKeachie, the "learning motivation" focused on in this study is divided into three dimensions: value, expectancy, and affect, with 31 items in total. The value dimension is further divided into intrinsic goals, extrinsic goals, and work value; the expectancy dimension includes learning and performance self-efficacy and control beliefs; and the affect dimension includes an anxiety assessment [53,54]. The version of the MSLQ used in this study is based on the revised version by Wu, J.J.; Cherng, B.L. [55] and refers to the revised version by Huang, Y.W. [56], covering the evaluation of the three major dimensions, value, expectancy, and affect [57]. The test was developed using a five-point Likert scale, with the item descriptions modified based on the context of the "2024 Taiwan Craft Academy summer talent training camp". Since this study aims to gain an in-depth understanding of students' expectations and motivations for participating in the "2024 Talent Training Camp", an interview outline was developed based on the dimensions of the MSLQ framework.

#### 3.3.2. Imagination Thinking Scale

The core objective of craft and design education is to cultivate talents who possess the abilities of future designers [58]. Craft and design education not only imparts professional knowledge and skills but also emphasizes the development of aesthetics, technology, science, integration, and creativity. Among these, the education of imagination uses thinking strategies to enhance students' thinking abilities, enriching their creativity. Interdisciplinary learning plays a crucial role in accumulating rich life experiences, which facilitates the extraction of data during imaginative thinking and broadens the scope of imagination, allowing creativity to be fully unleashed. The cultivation of imagination is an essential foundation in craft and design education. In recent years, the T-shaped talent theory has highlighted the importance of imagination, encouraging individuals to engage in dialogue across various abilities and interests and interacting with core competencies to create unique personal value. The greater the distance between the domains of interdisciplinary thinking, the more unique and innovative the integration ability becomes [59]. Vygotsky, on the other hand, believed that imagination is built upon accumulated memories from the past [60], and as it develops, it becomes a complex system that integrates language, logic, and perception, which is externally expressed through verbal communication and behavior [43].

In 2012, domestic scholars Tsau, S.Y., Lin, H.H. [61] published the "Imagination Power Scale", which was developed based on Vygotsky's concept of imagination. This scale includes four activity items and takes 40 min to complete. The content of the scale includes four aspects: richness of imagery (initiation), fluency, flexibility, and originality. The overall Cronbach's reliability coefficient of the scale is 0.77. The retest reliability, tested by correlation coefficients, was 0.71, which is statistically significant. Confirmatory factor analysis further confirmed that the scale achieves good convergence validity indicators. This makes it suitable for use in this study as an evaluation tool for the "2024 Taiwan Craft Academy summer talent training camp".

## 4. The Practice and Reflection of Craft Innovation Education

### *4.1. 2024 Taiwan Craft Academy Summer Talent Training Camp*

The 2024 Taiwan Craft Academy summer talent training camp, planned and promoted by the Craft Center, is based on the principle of craft innovation education. The focus is on the innovative interdisciplinary dual mentorship education strategy, which aligns with the "Creative Spiral Ascension Theory" proposed in the research. This theory emphasizes the upward elevation of creativity through interdisciplinary stimulation, where creative ascension is demonstrated in a bottom-up process—progressing from Qi-form (material) to Xin-form (psychological) and ultimately to Dao-form (philosophical) practice. The following explores the features of the four experimental lab courses, along with the analysis of participants' post-course MSLQ interviews.

#### 4.1.1. Lacquer Craft: Cross-Technical Integration and Innovation

1. **Course Practice Content:** The course emphasizes "cross-disciplinary techniques" in the lacquer craft, blending traditional techniques with innovative technologies. Traditional lacquer techniques such as varnishing, overlaying, mother-of-pearl inlay, eggshell attachment, and other classic lacquer processes focus on intricate manual skills and profound technical accumulation, which are becoming increasingly rare in the craft and art departments of universities. In terms of innovation, 3D modeling and printing technologies are integrated to transform the traditional forms of lacquer craft, aligning with modern small-scale production techniques. These techniques enhance the unique characteristics of artworks while breaking the limitations of material forms in shaping details and imagination. The use of 3D printing in lacquer craft not only accelerates the production process but also allows for the creation of complex shapes that traditional manual techniques cannot achieve, imbuing the pieces with modernity and unique stylistic qualities. The core of the "Lacquer Art 3D Jewelry Creation Camp" lies in breaking the boundaries of "single technical fields" and allowing creators to experiment more freely with different techniques. Through the interaction of various techniques, creators can reach a broader cross-disciplinary imagination based on their experiences.
2. **Analysis of Student Works:** The concept of "I Ching, Explaining the Hexagrams" reflects the awareness of natural laws, which include the study of the cycles of the four seasons and the harmony between day and night. From a creative perspective, this concept can be interpreted as the interactive relationship and aesthetic harmony

between color, material, technique, and concept. From the in-depth interviews, it was found that student Hong OO's work "Flower Dance–Bracelet 1", shown in Figure 5a, skillfully uses 3D printing to its advantage in shaping. The inspiration was drawn from the blooming and falling of flowers, symbolizing life's changes and cycles, representing the process of survival and transcendence. The work employs enamel techniques and natural lacquer to express the vitality of flowers, giving us a sense of the impermanence and beauty of nature. Student Liu OO's work, "Tunnel of Time", shown in Figure 5b, adopts a component-based assembly concept, blending tradition with modernity. The changing shapes, colors, and combinations represent the flow of time, as described in the *I Ching*'*s* Feng: "The day begins with sunrise and ends with sunset, the moon waxes and wanes, heaven and earth are full and empty, changing with time". This concept sends a message about the time changes in nature, guiding people to follow the rhythm of life and become a representation of the ongoing vitality of life. This is symbolized in the work's interconnected ring structure, representing the continuity and innovation of culture, reflecting the intertwined relationship between tradition and innovation in craft creation. The work Flowers Fly, Flowers Fall by participant Hong OO, as shown in Figure 5c, expresses the concept of Yin's emptiness and Yang's solidity by shaping the intangible form of wind. The surface incorporates the maki-e technique of lacquer craft, delicately depicting the dynamic posture of flowers dancing in the air as they are carried by the wind. The works Trajectory 12:00 and Trajectory 17:00 by participant Song OO, as shown in Figure 5d, demonstrate the creator's exploration of void and solidity, large and small, light and shadow through lacquer craftsmanship. These pieces reflect a meticulous observation of the changing natural forms under different solar trajectories, using gold leaf and lacquer's unique properties to convey the artist's perception of the environment. The work Wheel Giraffe by participant Liu OO, as shown in Figure 5e, embodies an anti-artistic concept, aiming to balance humor and critique. It intertwines the fragility of "existence" with the contradictions of technological development in "life", provoking reflections on environmental conservation and respect for life. The piece adopts the giraffe as a symbolic figure, gradually transforming from a realistic representation into an abstract symbol, illustrating humanity's excessive exploitation of nature. This process critiques the tendency to impose human societal meanings on natural entities, serving as a warning against the depletion of the world's resources. The work resonates with Eastern philosophy's emphasis on harmony with nature and sustainable development.

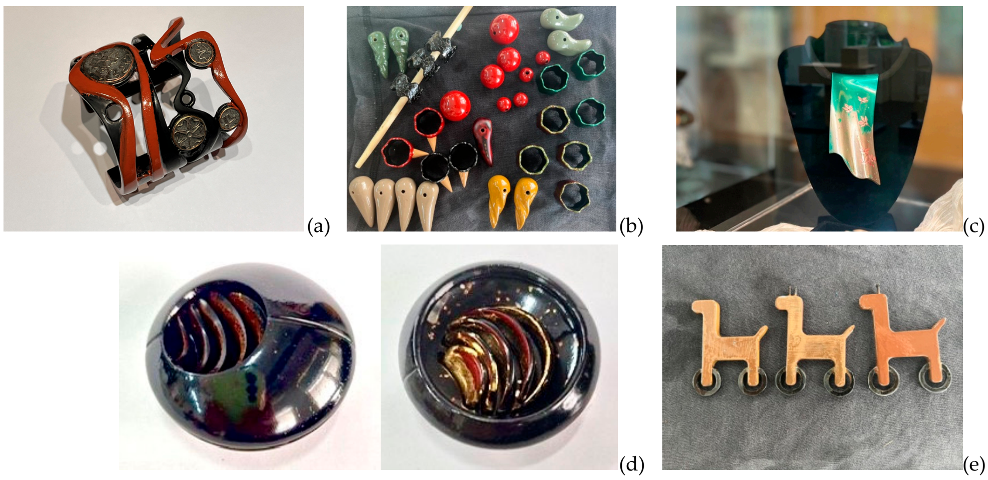


**Figure 5.** Lacquer craft laboratory student works: (**a**) Hong OO "Flower Dance—Bracelet 1"; (**b**) Liu OO "Tunnel of Time"; (**c**) Hong OO "Flowers Fly, Flowers Fall"; (**d**) Song OO, "Trajectory 12:00", "Trajectory 17:00"; (**e**) Liu OO "Rotating Deer".

#### 4.1.2. Woodcraft: Technological and Technical Advancements in Depth

1. **Course Practice Content:** The woodcraft course emphasizes the "interaction between deep technical skills and knowledge", training students to master and advance their technical abilities. The course is designed in response to the limited training available in university industrial design, crafts, and cultural industries programs, where due to constraints in faculty, time, and equipment, students only gain a superficial understanding of basic woodworking techniques. This gap between educational offerings and industry needs is significant. Therefore, the "Wood Craft Combined with Digital Manufacturing Technology Application Training Course" includes not only traditional woodworking techniques, such as the precise mastery of joinery, but also incorporates advanced technological applications, exploring how digital design and manufacturing techniques (such as CNC and 3D modeling software) can expand creative possibilities. Students learn to design furniture structures using computer software, understand the properties of wood materials and their application limitations, and operate digital manufacturing tools to complete practical projects. From a conceptual standpoint, depth cross-disciplinary learning emphasizes "comprehensive knowledge integration" and "systematic thinking". The course cultivates students' deep understanding of materials, structures, and technical processes while honing their operational skills.
2. **Analysis of Student Works:** The concept from Laozi's Tao Te Ching (Chapter 42), "All things carry Yin and embrace Yang; they are the blend of the two forces", suggests that natural phenomena inherently contain both Yin and Yang, which may seem contradictory but are essential for mutual existence. In other words, the interplay, clash, and harmony of Yin and Yang ultimately lead to a state of balance. This concept is notably reflected in the students' creative works in woodcraft. In the in-depth interviews, student Yang OO shared that his work "Return to Fun", shown in Figure 6a, was inspired by his childhood memories. He created a wooden tool cart symbolizing "fun and memories" in life, capturing intricate family interactions and familial love. The work integrates traditional joinery structures with modern CNC technology, embodying the functional value of an object within the realm of survival while infusing the creator's longing for innocence and childhood memories. This elevates the object, which exists only at the "form" level, to the "heart" level, reflecting continuous innovation and a return to the origins of life's wisdom. The aforementioned interplay of Yin and Yang, the visible conflict between extremes, is essential for harmony. This concept is particularly well-expressed in student Jiang OO's work "Camphor Suspension", shown in Figure 6b. In Kevin Jiang's conceptualization, this work explores the interaction between tensile structure and the flexibility of wood. The chair design demonstrates the tension and balance between Yin and Yang, as described in the *I Ching's* Xi Ci I: "One Yin and one Yang is the Dao, and those who follow it are good, those who achieve it are natural". This reflects that all things are born from the fusion and harmony of Yin and Yang, found through conflict and contrast. The work is based on the concept of opposing yet harmonious forces of heaven and earth, interpreting the fusion of science and craftsmanship through the visual representation of Tai Chi. In reality, the creative process of this work involved continuous experimentation and trial and error, ultimately achieving a perfect balance. Student Zheng OO, in Figure 6c, mentioned that his work "The Two Forms of Heaven and Earth" uses the core design principle of Tai Chi, interpreting the balance and opposition between heaven and earth. The work employs rosewood and walnut materials to embody the profound meaning of Yin and Yang harmony and the integration of hardness and softness, a key aspect of Eastern philosophy. The structure of the work aligns with the concept of "Dao form" and expresses the deep interrelation of Yin and Yang through its color and

shape. It links philosophy with craftsmanship, revealing the deeper life layer required in design. The use of Tai Chi as a symbol in this work reflects the creator's intent, illustrating the connection between form and meaning. This concept, mentioned in *I Ching's* Xi Ci I, explains, "The sage establishes symbols to express intention, creates hexagrams to embody feelings and falsehoods, and connects the text to its words". The relationship between form and meaning, expressed through symbols, is a common design approach used by contemporary designers, conveying subjective ideas through objective symbols—an aesthetic approach called "observing things to extract symbols" to establish the relationship between subjective and objective meanings. In Figure 6d, participant Guan OO's work, Pigment Cabinet, represents the creator's memories of their mother. Since the creator's mother was a painter, their home was always filled with various pigments stored in cabinets. Growing up alongside their mother's artistic practice, the creator incorporated playful geometric shapes into the cabinet's structural design. The pigment jars placed within the cabinet are arranged randomly, transforming it into a piece of furniture with a living story.

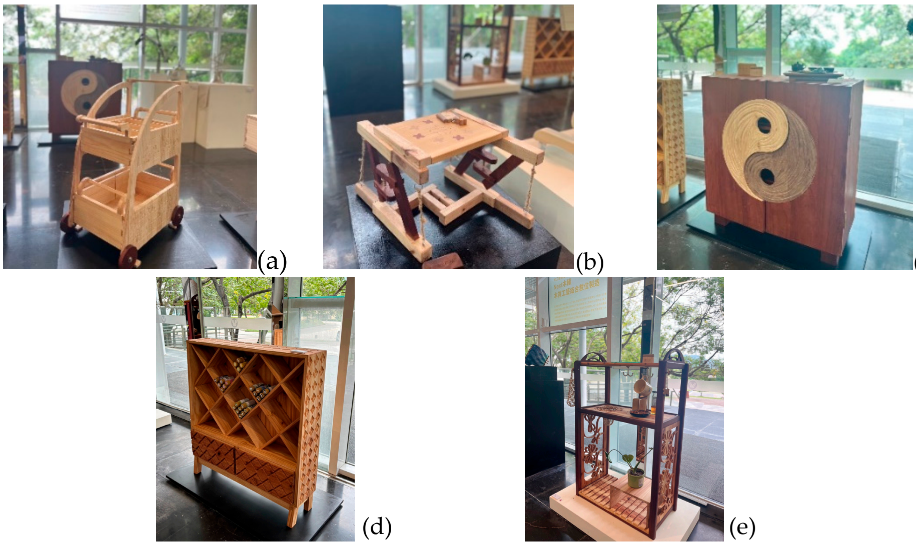

**Figure 6.** Woodcraft laboratory student works: (**a**) Yang OO "Return to Fun"; (**b**) Jiang OO "Camphor Suspension"; (**c**) Zheng OO "The Two Forms of Heaven and Earth"; (**d**) Guan OO Pigment Cabinet; (**e**) Huang OO Flower Window Delicacies.

Regarding emotional expression, Figure 6e features participant Huang OO's work, Flower Window Delicacies, which also draws inspiration from childhood memories. In the creator's home, there was always a cabinet filled with various snacks, while the wooden structure's side panels reflect memories of the plants and trees from childhood garden play. This nostalgic recollection is further emphasized in the naming of the work, playing on the homophonic connection between the Chinese characters for "time" (shí) and "food" (shí).

#### 4.1.3. Metal Crafts: Breaking Boundaries of Aesthetic Concepts Through Connection and Dialogue

1. **Course Practice Content:** The uniqueness of the metalcraft course lies in cross-disciplinary conceptual thinking, attempting to achieve a balance between traditional tin craftsmanship and modern craft aesthetics for contemporary life. Its distinctiveness is embodied in the "connection and dialogue between technique, aesthetics, and concept". The core of the course is to explore the integration of tin crafts and floral design. Students will learn the basic material handling and techniques of tin and participate in practical courses such as three-dimensional modeling, welding, and casting. Addi-

tionally, from the perspective of floral design, the course explores consumer profiles, lifestyles, and aesthetics, prompting students to think about the symbolic meaning and aesthetic value of their works. The cross-disciplinary craftsmanship is elevated to the "conceptual dialogue" level, allowing traditional techniques to become not only tools to achieve specific shapes but also carriers of design concepts and life aesthetics. This facilitates a dialogue between the work and its users, potentially leading to a change in mood or behavior. It is also important to note that both tin crafts experts and floral designers agree that such creations must be highly planned, as the work serves as the medium of dialogue with the consumer. Therefore, the dual mentorship system plays a critical role, guiding students through the process of translating abstract ideas into tangible works, which communicate emotions, thoughts, or concepts.

2. **Analysis of Student Works:** According to the *I Ching's* Kun Hexagram, "The noble person is centered in the Yellow, aligning with the Way. When he stands in the proper position, beauty flows from within and extends outward, manifesting in the four limbs and progressing in the affairs of life—this is the highest form of beauty". This describes a noble person's internal sincerity and harmony, and how this cultivates virtue and beauty in life, manifesting in both "beauty" and "goodness". It also reflects a return to nature and an alignment with one's true self, achieving the realm of beauty. Because the metalcraft course combines tin craftsmanship with floral design, students express their understanding of the Dao through their creative works. Student Wu OO, in an in-depth interview, mentioned that his work, Untitled, shown in Figure 7a, uses tin as a frame to symbolize the changes and transformations of time and people. The seemingly contradictory use of tin's hardness and the softness of flowers, however, displays the harmonious cycle of life through the fusion of Yin and Yang, illustrating the principle of "life evolves in the I Ching", merging "form" and "heart" in craftsmanship. Student Chen OO's work "Dropping Rhythm", shown in Figure 7b, was inspired by the shape of a water droplet, symbolizing the origin of all life. The design uses the malleability and purification function of tin to express the flow and creation of life through curves, in line with the concept of "Dao follows nature". The work is not just a flower vase but also reflects the creator's understanding of natural life, transforming the "Dao form" into a tangible expression, exemplifying the practice of the "creativity spiral sublimation theory". Student Lin OO's work "Seed Stone", shown in Figure 7c, combines tin and plants to express vitality and resilience in a challenging environment. The work explores the transformation of Yin and Yang and the concept of life. As described in the *I Ching's* Xi Ci I, "The opening and closing are called change; the back-and-forth is called flow"; it illustrates the movement of "Qi" in nature, characterized by an adaptability to time and place. Lin OO's work shows how life can express itself according to external changes while maintaining internal harmony, reflecting the wisdom of the dynamic flow of Qi and time in nature. As stated in the perspective of the *I Ching's* Xi Ci I (*Book of Changes*, Commentary on the Appended Phrases, Part One), student Lin OO's work, Flow, Figure 7d, demonstrates exquisite pewter craftsmanship, forging metal into the form of flowing liquid. This design captures the dynamic movement of water, breaking away from the conventional structure of vases. The work symbolizes the idea that the infinite flow of water nourishes all living things, naturally giving rise to plant life.
3. Lastly, student Chen OO's work Flowers and Vase, Figure 7e, is inspired by the purifying properties of pewter in water. The artist meticulously crafted a pewter rose, breaking away from the traditional appearance of vases. When real flowers are placed in the piece, it creates a harmonious blend of the tangible and intangible,

introducing an asymmetrical aesthetic that redefines the conventional concept of floral arrangements.

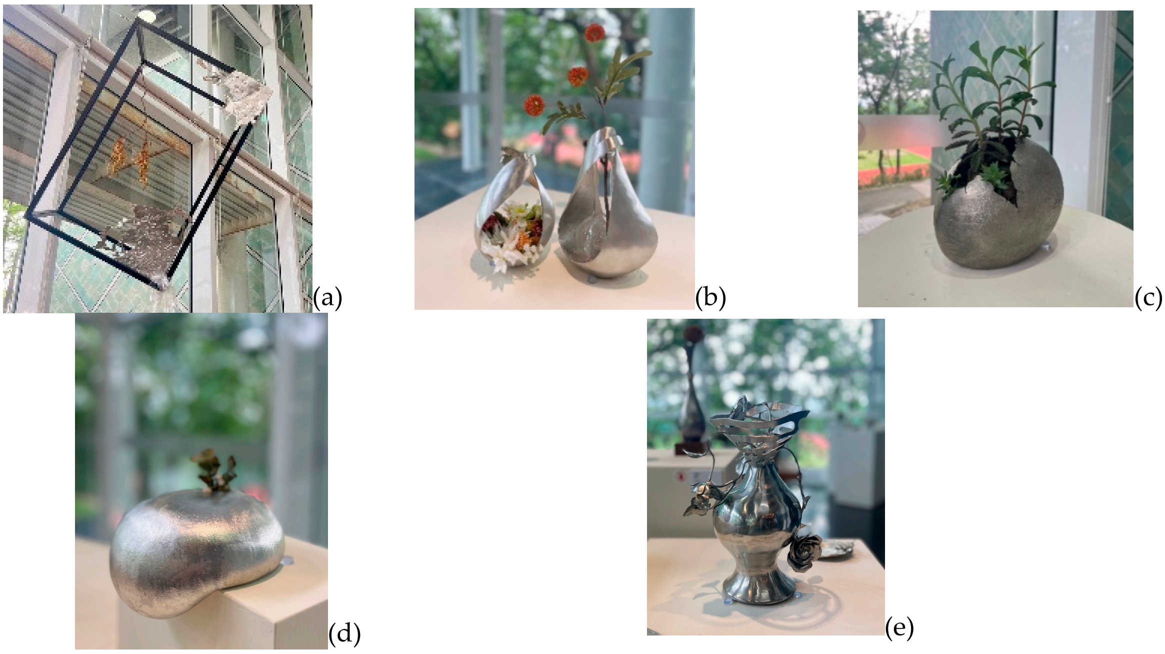

**Figure 7.** Metalcraft laboratory participants' works: (**a**) Wu OO, "Untitled"; (**b**) Chen OO, "Dripping Rhythm"; (**c**) Lin OO, "Seed Stone"; (**d**) Lin OO Flow; (**e**) Chen OO Flowers and Vase.

#### 4.1.4. Ceramic Arts: Integration and Fusion of Cross-Materials

1. **Course Practice Content:** The ceramic arts course is designed with ceramics at the core and incorporates other materials as complementary elements, focusing on the "cross-disciplinary fusion of different material characteristics" and the "interaction between technique, design, and materials" in both teaching and creative exploration. The course, titled "Composite Ceramic Sound System Integrated with 3D Printing Design", not only focuses on ceramic manufacturing techniques but also integrates 3D printing technology and other materials, such as wood, allowing students to explore new materials and technologies in the application of ceramics. This approach to cross-material integration requires students to master the characteristics and application domains of multiple materials, and to merge them within the limits of each material. This includes understanding ceramics' plasticity, the changes that occur during the firing process, the precision of 3D printing technology, and the application of wood in structural design. More specifically, the course covers 3D printing, sound structure design, woodworking, glaze formulation, bisque firing, and other techniques. The objective of this cross-material teaching is to cultivate students' innovative abilities to "break material boundaries".
2. **Analysis of Student Works:** The interaction between glaze colors and temperature in ceramic arts, as well as the integration of cross-disciplinary materials, opens up the creative imagination of students. The philosophy of the *I Ching* echoes this creative process, especially the concept of aesthetic craftsmanship. As stated in the *I Ching's* Xi Ci II, "The interaction of hardness and softness brings change", which emphasizes the fusion of Yin and Yang in everything, seeking a balance where seemingly stillness embodies the origins of life, a principle that also stimulates creative energy. As shown in Figure 8a, student Zhang OO's work Swaying demonstrates the fusion of ceramics and wood, using unstable shapes to create a dynamic and interactive feel in the piece. It symbolizes an unshakable will swaying in the wind. The design

not only reflects resilience and perseverance in life but also expresses philosophical thinking on Yin–Yang balance through the contrasting dark and light glazes. Student Hong OO, in an interview, mentioned that her work Ultramarine (Figure 8b) was inspired by her own experiences and the mountains and rivers of her hometown in Nantou. The piece uses blue ceramics to depict the flowing atmosphere of the natural landscape, illustrating the interaction between nature and life. Her work conveys a keen awareness of life aesthetics, which also aligns with the philosophical understanding of creativity, echoing the creation philosophy of "life is the Dao". Through her acute sensitivity to natural phenomena, she transforms them into art, interpreting the flow and balance of life. Another notable work is Solar Eclipse by student Guo OO (Figure 8c), which transforms the astronomical phenomenon of a solar eclipse into a dual sensory experience of both visual and auditory art. The work uses a combination of different materials to express creativity. The ceramic trumpet form in the piece represents the solar eclipse, with the light and dark changes symbolizing transformation and constancy, while the colors reflect the philosophical concept of Yin–Yang fusion. The work aims to provide the viewer with a multisensory experience in both sight and sound, allowing them to understand the natural laws and cosmic operations just as the artist does. In the interdisciplinary material integration creation, Ruan OO also delivered an outstanding performance with his work Phantom (Figure 8d). He boldly experimented with various materials, including ceramics, teakwood, and electronic components, drawing inspiration from the "blessing horn" found in Taiwan's indigenous traditional culture. By replacing the traditional animal horn material, originally obtained through hunting, with ceramics, his design not only exudes a contemporary aesthetic but also integrates the concept of blessings into the design process. This allows users to experience a sense of blessing while listening to music through the speaker in daily life. In Wu OO's work Nature's Echo (Figure 8e), the artist took inspiration from the ecological textures of the natural monstera plant. The unique shape and patterns of its leaves symbolize the power and harmony of nature. By utilizing ceramic materials, the piece enhances sound clarity, further reinforcing the connection between nature and auditory experience.

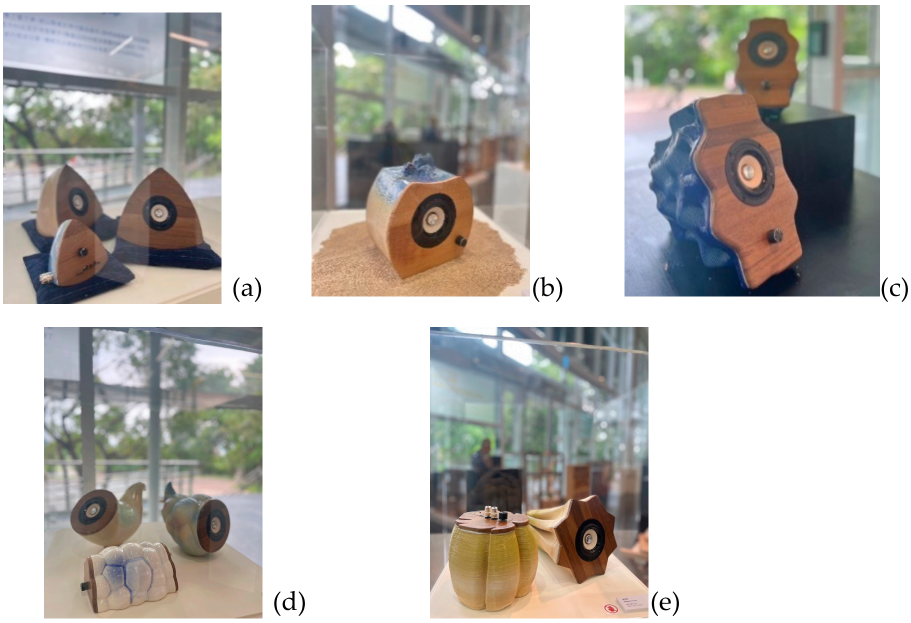

**Figure 8.** Ceramic arts laboratory student works: (**a**) Zhang OO "Swaying"; (**b**) Hong OO "Ultramarine"; (**c**) Guo OO "Solar Eclipse; (**d**) Ruan OO "Phantom"; (**e**) Wu OO "Nature's Echo".

## 4.2. Review and Reflection on Educational Practice

### 4.2.1. Impact of Cross-Disciplinary Education Strategies on Students' Imagination Performance

The "2024 Taiwan Craft Academy summer talent training camp" implemented different cross-disciplinary educational strategies across its four laboratories. These strategies included technical cross-disciplinary in the lacquer craft, deep cross-disciplinary in woodworking, conceptual cross-disciplinary in metalworking, and material cross-disciplinary in ceramic arts. Each laboratory adopted a multi-mentor teaching model. Although the number of participants varied slightly due to limitations in recruitment, space, equipment, and resources, the students' performances still provide valuable reference points.

Therefore, this study chose to use one-way ANOVA (analysis of variance) to examine whether there were inter-group differences in the students' imagination performance, including aspects such as "originality", "fluency", "flexibility", and "uniqueness" after their training in the camp. The results showed statistically significant differences across the various aspects of imagination among students from different teams (Table 2).

**Table 2.** Summary of variance analysis of imagination ability among participants from different camps.

| | | Sum of Squares | df | Mean Square | F | Significance |
|---|---|---|---|---|---|---|
| Originality | Between Groups | 89.415 | 3 | 29.805 | 9.903 | 0.000 |
| | Within Groups | 117.376 | 39 | 3.010 | | |
| | Total | 206.791 | 42 | | | |
| Fluency | Between Groups | 53.160 | 3 | 17.720 | 9.104 | 0.000 |
| | Within Groups | 75.909 | 39 | 1.946 | | |
| | Total | 129.070 | 42 | | | |
| Flexibility | Between Groups | 56.129 | 3 | 18.710 | 4.040 | 0.014 |
| | Within Groups | 180.615 | 39 | 4.631 | | |
| | Total | 236.744 | 42 | | | |
| Innovativeness | Between Groups | 28.309 | 3 | 9.436 | 3.209 | 0.033 |
| | Within Groups | 114.668 | 39 | 2.940 | | |
| | Total | 142.977 | 42 | | | |

1. Originality Performance: The post hoc Duncan test (Table 3) showed significant differences in the "originality" dimension of imagination among the students from different camps ($p$ = 0.000). Notably, the woodworking camp demonstrated exceptional performance. This study suggests that the cross-disciplinary nature of woodworking, with its focus on deep technical integration, allowed students to explore techniques in greater depth. This addressed the current limitations in resources, space, and equipment in higher education institutions, providing students with breakthrough creative support based on their prior learning experiences. This aligns with Vygotsky's [30] theory of learning, which suggests that students' imaginative performance originates from their accumulated experiences, and through the provision of deep technical knowledge and equipment, students are offered critical opportunities to reframe their experiences and spark creativity, serving as material for conceptualizing ideas.
2. Fluency Performance: The post hoc Duncan test results for the "fluency" dimension of imagination (Table 4) revealed that the four camps could be broadly divided into three groups. Notably, the metalworking students performed the best in fluency, followed by the ceramic arts students (Table 4). Fluency refers to the sensitivity of students to the idea-generation process, reflected in the number of creative associations made. In-depth interviews with metalworking students indicated that the instructors emphasized continuity and logic in the creative and design processes, which facilitated more planned production. As a result, students interacted closely with the instructors

during the ideation phase and quickly generated creative solutions, maintaining continuity in the design process. In other words, the metalworking process demands high precision and technical skill and is less amenable to post-production adjustments, prompting students to focus more on early-stage planning, thereby enhancing their fluency in the imaginative process.

**Table 3.** Duncan post hoc test for the originality dimension.

| | | Alpha = 0.05 | |
|---|---|---|---|
| Training camp | N | 1 | 2 |
| Ceramic Arts | 9 | 3.44 | |
| Lacquer Craft | 11 | 4.64 | |
| Metal Crafts | 15 | 4.87 | |
| Woodcraft | 8 | | 7.88 |
| Significance | | 0.088 | 1.000 |

**Table 4.** Duncan post hoc test for the fluency dimension.

| | | Alpha = 0.05 | | |
|---|---|---|---|---|
| Training camp | N | 1 | 2 | 3 |
| Lacquer Craft | 11 | 6.64 | | |
| Woodcraft | 8 | 7.88 | 7.88 | |
| Ceramic Arts | 9 | | 8.89 | 8.89 |
| Metal Crafts | 15 | | | 9.40 |
| Significance | | 0.052 | 0.109 | 0.414 |

3. Flexibility Performance: Duncan's post hoc test analysis for the "flexibility" dimension showed significant differences between camps ($p$ = 0.014), indicating varied effects of the training on students' flexibility development. The ceramic arts students demonstrated significantly higher flexibility than those in other courses after training (Table 5). This result is likely due to the ceramic arts course's emphasis on using a variety of materials, combining 3D printing technology with traditional kiln firing. The fusion and clash between different materials, technology, and tradition helped students develop flexibility in applying various materials and harmonizing them, showcasing their ability to draw out the unique features of each material and express their creative concepts. This approach broke the constraints of using a single material, allowing students to develop superior flexibility in their creative process.

**Table 5.** Duncan post hoc test for the flexibility dimension.

| | | Alpha = 0.05 | |
|---|---|---|---|
| Training camp | N | 1 | 2 |
| Lacquer Craft | 9 | 5.27 | |
| Metal Crafts | 11 | 5.93 | |
| Woodcraft | 15 | 7.25 | 7.25 |
| Ceramic Arts | 8 | | 8.33 |
| Significance | | 0.056 | 0.263 |

4. Uniqueness Performance: The post hoc Duncan test for the "uniqueness" dimension revealed significant differences between camps ($p$ = 0.033), with the woodworking students showing the most outstanding performance, followed by the lacquer arts students (Table 6). This study suggests that the woodworking course provided students

with more advanced equipment, enabling them to break free from the conventional thinking shaped by past experiences and limited by equipment. This allowed them to explore new creative methods beyond traditional concepts and material applications. Furthermore, woodworking materials are highly diverse and malleable, offering greater creative potential. Techniques such as carving, joining, sanding, and coloring, as well as the use of digital technology, enabled students to make bold attempts and innovate, generating creative solutions quickly and fostering a “creation from nothing” mindset that enhanced their ability to think independently and uniquely.

**Table 6.** Duncan post hoc test for the innovativeness dimension.

| | | **Alpha = 0.05** | |
|---|---|---|---|
| Training camp | N | 1 | 2 |
| Metal Crafts | 15 | 6.20 | |
| Ceramic Arts | 9 | 6.56 | |
| Lacquer Craft | 11 | 7.64 | 7.64 |
| Woodcraft | 8 | | 8.25 |
| Significance | | 0.081 | 0.425 |

#### 4.2.2. The Relationship Between Learners’ Motivation and Imagination Learning Outcomes

Previous research has pointed out that learners’ motivations significantly influence learning outcomes through factors such as internal control, learning strategy application, and attitudes toward challenges [43]. To examine whether the results of the “2024 Taiwan Craft Academy summer talent training camp” align with past research, this study will use Pearson’s correlation analysis to investigate the relationship between learners’ motivation and their imagination performance after training.

Table 7 shows that the value factor of learning motivation is significantly correlated with both originality and fluency. Specifically, there is a strong correlation between learners’ “intrinsic goals” and “originality”, with a correlation coefficient of 0.822. This suggests that when learners are highly motivated by intrinsic goals, such as a strong desire to be proactive, inquisitive, and engaged in the camp, it positively influences their performance in the “originality” dimension of imagination. Additionally, the “work value” factor also shows a significant correlation with “fluency”, with a correlation coefficient of 0.329. This indicates that when learners perceive the training content as beneficial for their future, life, work, and learning processes and find it meaningful and valuable, it stimulates them to think quickly and freely, thereby enhancing their fluency in imaginative thinking.

In terms of the relationship between the expectancy factor of learning motivation and imagination, Table 7 shows that “self-efficacy” is significantly correlated with both “fluency” (correlation coefficient = 0.380) and “uniqueness” (correlation coefficient = 0.330). According to Bandura [62], self-efficacy refers to learners’ confidence in their learning abilities, especially in facing difficulties and challenges. Learners who are confident in their ability to manage and complete tasks will perform better in imaginative thinking tests, demonstrating greater fluency and originality. The “expectation of success” factor is highly significantly correlated with “uniqueness” (correlation coefficient = 0.883). This is consistent with Pintrich’s view that the strength of learners’ beliefs in their ability to succeed and reach their goals is a significant factor influencing learning outcomes [63]. In the context of the “2024 Taiwan Craft Academy summer talent training camp”, the learning motivation factor “expectation of success” is significantly associated with learners’ “uniqueness” in imagination.

**Table 7.** Summary of the correlation analysis between learning motivation and imagination.

| | | Value | | | Expectation | | | Emotion |
|---|---|---|---|---|---|---|---|---|
| | | Intrinsic Goal | Extrinsic Goal | Work Value | Control Belief | Self-Efficacy | Expectation of Success | Emotionalization |
| Originality | Pearson correlation | 0.822 ** | 0.149 | −0.108 | −0.072 | 0.065 | 0.100 | −0.477 ** |
| | Significance | 0.000 | 0.339 | 0.490 | 0.648 | 0.679 | 0.522 | 0.001 |
| | N | 43 | 43 | 43 | 43 | 43 | 43 | 43 |
| Fluency | Pearson correlation | 0.218 | −0.064 | 0.329 * | 0.192 | 0.380 * | −0.157 | −0.356 * |
| | Significance | 0.161 | 0.682 | 0.031 | 0.218 | 0.012 | 0.314 | 0.019 |
| | N | 43 | 43 | 43 | 43 | 43 | 43 | 43 |
| Flexibility | Pearson correlation | 0.138 | 0.123 | −0.099 | −0.092 | 0.147 | 0.155 | −0.420 ** |
| | Significance | 0.377 | 0.434 | 0.527 | 0.556 | 0.346 | 0.320 | 0.005 |
| | N | 43 | 43 | 43 | 43 | 43 | 43 | 43 |
| Innovativeness | Pearson correlation | 0.156 | 0.105 | 0.223 | 0.007 | 0.330* | 0.883 ** | −0.471 ** |
| | Significance | 0.319 | 0.504 | 0.150 | 0.966 | 0.030 | 0.000 | 0.001 |
| | N | 43 | 43 | 43 | 43 | 43 | 43 | 43 |

* $p < 0.05$; ** $p < 0.01$.

It is worth noting that in Pintrich's learning motivation strategy theory, the emotional factor is defined as a learner's anxiety regarding anticipated tests, exams, or presentations. In the face of such pressures, learners may experience negative thoughts due to concerns about poor performance or may have physiological and emotional reactions related to stress. In fact, during the in-depth interviews with the learners, this emotional factor caused difficulties for some students, particularly as all the camps planned joint outcome presentations and exhibitions. This led to situations where learners felt tense, unable to concentrate, or struggled to focus on their creative work during the creation process or in interactions with other students. This outcome is also reflected in the correlation analysis between learning motivation and imagination.

Table 7 clearly shows that there is a significant negative correlation between the "emotional factor" of learning motivation and all aspects of imagination. This means that as learners' anxiety and pressure about post-event exhibitions and presentations increase, their performance in various imagination dimensions is significantly suppressed. This result aligns with Pintrich et al.'s [52] learning motivation strategy theory, which suggests that excessive emotional pressure leads to learning interference and discomfort, demonstrating a negative relationship between the two. In other words, in the creative process of the "2024 Taiwan Craft Academy summer talent training camp", emotional factors inhibited imagination performance. It is crucial for future course planning to consider how teachers can guide learners with the right attitude and appropriate learning pressure to facilitate effective teaching.

## 5. Conclusions and Recommendations

### *5.1. Insights from Eastern Philosophy on Craft Education*

This study begins with a review of Eastern philosophical thought to explore the possibilities of contemporary craft innovation education, using the "2024 Taiwan Craft Academy summer talent training camp" organized by the Craft Center as a case study in educational practice. This study primarily discusses the ideas from Laozi's *Tao Te Ching* and the *I Ching* while also referencing the "Three-Life Needs Theory" proposed by Chen, T.L. and Zhuang, Y. [1]. From the perspective of Taiji (the Supreme Ultimate), this study discusses how craft creation and education, like the cycle of life in all things, possess the qualities of cross-domain integration, coordination, and balance. Within the regulation of change and permanence, they form the driving force for the evolution of life. This study presents the "Cross-Domain and Harmonious Relationship Diagram" and concludes with the concept of imagination from the educational theorist [52], which holds that human creative activities originate from the accumulation, recombination, and transformation of past experiences. Based on the "Spiral Innovation Theory" proposed by Chen, T.L., Hong, Y.H., and Lee, Y.C. (2023) [48], this theory is grounded in the Taiji harmony of Yin and Yang and describes the upward internal sublimation process of the creator's form, heart, and Tao. This process represents a moral realization that is manifested in the work, which becomes an expression of creativity and imagination.

### *5.2. Educational Practice Strategy*

As emphasized by Kolb's [42] experiential learning theory, students should not merely be passive receivers of information but must develop the ability to understand knowledge applications through bodily sensations, experiences, reflections, and the abstraction of concepts. This is the key feature of craft innovation education, which also offers a balancing opportunity for contemporary education. The learning process for students involves an upward journey of understanding the principles of shape, mind, and Dao, gradually internalizing and constructing their own knowledge system. Based on Eastern philosophical discourse and its application in contemporary craft innovation education, this study attempts to use the "multi-mentor system" as an innovative educational strategy. This was implemented in the "2024 Taiwan Craft Academy summer talent training camp" organized by the Craft Center. The training camp consisted of four different craft-oriented laboratory categories—lacquer, woodworking, metalworking, and pottery—each with expert mentors from various fields developing interdisciplinary curricula with distinctive teaching features. In-depth interviews with the students revealed how these interdisciplinary courses stimulated creative imagination during the crafting process. This study further analyzed the results using the "Learning Motivation Strategy Scale" and "Imagination Thinking Scale", revealing that different interdisciplinary, innovative teaching strategies effectively influenced various aspects of imagination. For example, the woodworking course promoted idea generation, the metalworking course enhanced fluency, the pottery course improved adaptability, and the woodworking and lacquer courses contributed to originality. These results demonstrate how craft education, by inspiring imagination, helps to address the contemporary educational issue of overly focusing on industry demands, which can lead to a lack of depth in creative development and cultural reflection among learners.

### *5.3. Future Research Directions and Recommendations*

In the face of rapid industry changes, the educational model has shifted from the traditional "master-apprentice system" to the modern, systematized "learning system" model [64]. As the trend of craft-based teaching gradually declines, scholars such as Jehng et al. [65] remind us that within the context of learning, many educational programs fall under the "soft" fields (i.e., social sciences and arts/humanities). These programs, compared to those in the "hard" fields (i.e., engineering and business), tend to embrace uncertainty in the transmission of knowledge. Learners in these "soft" fields must engage in exploration, experience, and hands-on practice to achieve a complete learning experience. As emphasized by Kolb's [42] experiential learning theory, students should not merely be passive recipients of information but must understand the application of knowledge through bodily sensations, experiences, reflections, and abstractions. The expressions from contemporary educators resonate with the idea in this study that craft education should involve the upward realization of moral processes in form, heart, and Tao, internalizing and constructing one's own knowledge system. For practical cross-domain learning, this study suggests that the "dual-mentor system" strategy could be a key approach for future innovative education. By involving mentors from different fields, students can engage in debates, inquiries, and development in the classroom, mirroring the constant interaction in Taiji. More importantly, education should not be limited to the technical transmission of "form" skills; mentors in the master–apprentice system also serve as "role models", providing valuable resources for students' career development [66].

Overall, Eastern philosophy offers significant insights into contemporary craft innovation education. In response to sustainable challenges, cross-domain education strategies should be given more attention, and the dual-mentor system may serve as an effective framework for educational planning. This approach can enhance the teaching effectiveness of students' learning motivations and imagination, and these related topics deserve further research and consideration in the fields of craft innovation education and practical teaching. Furthermore, any researcher should revisit intangible cultural heritage from different cultures, especially reflecting on traditional philosophical thoughts that are on the brink of being forgotten, exploring their applicability and challenges within various cultural contexts. Based on this study, Eastern philosophical ideas integrated into contemporary craft innovation education should be subject to larger-scale empirical research. [67]. The course design should emphasize training in philosophical reflection. Exploring the effects of the "dual mentor system" strategy proposed in this study within contemporary educational environments will contribute to establishing a cross-disciplinary innovation education model that is both universally applicable and highly adaptable.

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